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Steps to diagonalize an $n \times n$ matrix $\mathbf{A}$:

Step 1. Find the eigenvalues and eigenvectors of $\mathbf{A}$.

Step 2. Check that there are enough linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ to form a basis of $\mathbb{R}^n$ (or $\mathbb{C}^n$).
If yes, set

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$
$$\mathbf{S} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{pmatrix} \quad \text{where } \mathbf{v}_i \text{ corresponds to } \lambda_i.$$

Important: The eigenvectors must be listed in the same order as their eigenvalues.
If any of the eigenspaces is deficient, there will not be enough linearly independent eigenvectors, and $\mathbf{A}$ is **not** diagonalizable.
Step 3. . Write $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.

Example 15.1 Let us diagonalize the matrix $\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$.

Step 1. The eigenvalues and eigenvectors of $\mathbf{A}$:

Eigenvalue Eigenvectors (basis of eigenspace) (3.33)

$$\lambda = 0; \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{3.34}$$

$$\lambda = -2; \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{3.35}$$

Step 2. The eigenvalues are of multiplicity 1, so the matrix is complete. Indeed, the two eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ form a basis of $\mathbb{R}^2$. Set

$$\mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \tag{3.36}$$

$$\mathbf{S} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \tag{3.37}$$

where the first column of $\mathbf{S}$ is the eigenvector of the eigenvalue 0 , listed in the first diagonal entry of $\mathbf{D}$, and the second column of $\mathbf{S}$ is the eigenvector of -2 , listed in the second diagonal entry in $\mathbf{D}$.

Step 3. Write $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$. If desired, compute $\mathbf{S}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ to get the explicit formula:

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1} \tag{3.38}$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{pmatrix}. \tag{3.39}$$

Problem 15.2 Diagonalize $\mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$.

Solution

We have previously found the eigenvalues and eigenvectors of $\mathbf{A}$:

eigenvalue eigenvectors (basis of eigenspace)

$$\lambda = 0; \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda = -3; \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Since each eigenspace is complete and there are 3 linearly independent eigenvectors, $\mathbf{A}$ is diagonalizable:

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1} \quad \text{where } \mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \end{pmatrix} \tag{3.40}$$

$$\mathbf{S} = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

(3.41)

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Concept check 1

1/1 point (graded)
Suppose that **S** is a matrix each one of whose columns is an eigenvector of a square matrix **A**. Assume also that **S** is square. Now, if the columns of **S** are linearly independent, then:

☐ **A** is invertible

☒ **S** is invertible

☒ **A** is diagonalizable

☐ **S** is diagonalizable



Solution:

If the columns of **S** are independent, then it has full rank, hence it is invertible. Also, if **S** is invertible, then the matrix **A** has a diagonalization.

The matrix **A** need not be invertible. Consider the matrix $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ whose eigenvalues are **0** and **1**, and eigenvectors are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively. The matrix **S** is square with linearly independent columns, but the matrix **A** has determinant **0** so has no inverse.

The matrix **S** need not be diagonalizable. We construct a matrix **A** that is diagonalizable via a matrix **S** which is not. Consider the matrix

$$\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The columns are linearly independent, but this matrix is not diagonalizable. It has a repeated eigenvalue of **1**, and a one-dimensional eigenspace. However, the matrix

$$\mathbf{A} = \mathbf{S} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{S}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

is a diagonalizable matrix with a matrix **S** that is not diagonalizable.

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You have used 1 of 3 attempts

Answers are displayed within the problem

Concept check 2

1/1 point (graded)
If the eigenvalues of **A** are **2**, **2**, and **5**, then **A** is:

☒ invertible

☐ diagonalizable

☐ not diagonalizable



Solution:

The determinant of **A** is the product of the eigenvalues, which is not equal to zero. Thus **A** is invertible. However, we do not know if **A** is diagonalizable or not because there is a repeated eigenvalue. If the eigenspace is 2 dimensional, then **A** is diagonalizable; otherwise **A** is not.

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Concept check 3

1/1 point (graded)

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no diagonalization



Solution:

The matrix **A** must have a repeated eigenvalue, and a deficient eigenspace. This implies that there is no diagonalization.

The matrix **A** may have an inverse. Consider the matrix $\begin{pmatrix} 3 & -0.5 \\ 8 & -1 \end{pmatrix}$, which is deficient with eigenvector $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$. This matrix is invertible with inverse $\begin{pmatrix} -1 & 0.5 \\ -8 & 3 \end{pmatrix}$.

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15. Examples of diagonalization

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